6 **(a)** No photoelectrons are emitted (from the metal surface) below the threshold frequency, f_0 regardless of the light intensity.

Electromagnetic radiation consists of photons with energy $E = hf$ in a 1-to-1 interaction with electrons, so a single photon with energy below hf_0 lacks the minimum energy required to free an electron.

(b) The emission line spectrum consists of a series of distinct lines of specific colours/ wavelengths/ frequencies on a dark background.

Each line is produced by the emission of photons of a specific quantum of energy. Since the photons have specific energy, they must be emitted when electrons in the atoms de-excite from a higher to lower discrete energy level.